



Aror University of Art, Architecture, Design and Heritage

Faculty of Artificial Intelligence and Multimedia and Gaming

Work Sheet: 01

Q:01 Find distance between two points and also find mid-point of the line segment joining the points.

- | | |
|---------------------------|---------------------------|
| (a) A(2, 3) and B(7, 11) | (b) C(-1, -4) and D(3, 1) |
| (c) E(0, 0) and F(6, 8) | (d) G(5, -2) and H(5, 4) |
| (e) I(-3, 7) and J(4, -1) | (f) X(0, 0) and Y(4, -1) |
| (g) I(4, 0) and Q(1, 1) | |

Q:02 Prove that the three points P(3,1), Q(6,0) and R (4,4) are the vertices of a right angle triangle?

Q:03 Sketch the graphs of the following equations, also find X-intercept and Y-intercept of each equation given below.

- | | |
|-------------------|---------------------|
| (a) $x + y = 4$ | (b) $x + 3y = 6$ |
| (c) $4x - 5y = 6$ | (d) $y = x - 8$ |
| (e) $y^2 = 4x$ | (f) $x^2 + y^2 = 4$ |

Q:04 Use the algebraic tests to check for symmetry with respect to both axes and the origin.

- | | |
|---------------|-------------------------|
| (a) $y = x^3$ | (b) $y = x^4 - x^2 + 3$ |
| (c) $xy = 4$ | (d) $x^2 - y = 0$ |

Q:05 Find the slope of the line passing through each pair of points.

- | | |
|---------------------------|---------------------------|
| (a) A(2, 3) and B(7, 11) | (b) C(-1, -4) and D(3, 1) |
| (c) E(0, 0) and F(6, 8) | (d) G(5, -2) and H(5, 4) |
| (e) I(-3, 7) and J(4, -1) | (f) X(0, 0) and Y(4, -1) |
| (g) I(4, 0) and Q(1, 1) | |

Q:06 Find the slope and y-intercept of the line.

(a) $x + y = 4$

(b) $x + 3y = 6$

(c) $4x - 5y = 6$

(d) $y = x - 8$

Q:07 Find the slope-intercept form of the equation of the line that has the given slope and passes through the given point.

(a) $m = 3, (0, -2)$

(b) $m = 4, (0, 0)$

(b) $m = -1/2, (2, -3)$

(d) $m = 6, (2, 3/2)$

Q:08 Determine whether the lines are parallel, perpendicular, or neither.

(a) $2x + 3y = 9$

(b) $x + 3y = 6$

$4x + 6y = 5$

$3x - y = 2$

(c) $x - 2y = 1$

$3x + 2y + 8$